Gradient

$f(x_1,\dots,x_n) = y$ Real-valued Multivariable function

f is differentiable if $\frac{\partial f}{\partial x_i} = \lim_{h\to 0} \frac{f(x_1,\dots,x_i+h,\dots,x_n) - f(x_1,\dots,x_n)}{h}$ exist $\forall i$

(shift i^{th} input by h)

We say f is continuously differentiable if a) f is differentiable ($\frac{\partial f}{\partial x_i}$ exist $\forall i \le n$)

We say f is $C^1(\mathbb{R})$. b) each $\frac{\partial f}{\partial x_i}$ is continuous on the domain of f.

$C^0(\mathbb{R})$ means continuous functions

$C^1(\mathbb{R})$ means differentiable and derivatives are $C^0(\mathbb{R})$

$C^n(\mathbb{R})$ means differentiable and all derivatives are $C^{n-1}(\mathbb{R})$

$C^\infty(\mathbb{R})$ means f is $C^n(\mathbb{R})$ $\forall n \in \mathbb{N}$ "smooth"

if differentiable then continuous

f, g, h $\quad f' = g \quad g' = h$

$f \xrightarrow{\frac{d}{dx}} g \xrightarrow{\frac{d}{dx}} h$

Lets say h is continuous. Then h is C^0.

Consider g. g is differentiable and $g' = h$ is C^0

↳ So g is C^1

Consider f. f is differentiable.

$f' = g$ is $C^1\Rightarrow f$ is C^2

↳ Hessian of f is symmetric $\iff f$ is C^2

- f is twice-diff AND $\frac{\partial f}{\partial x_i}$ is differen AND $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are all continuous

Gradient vector $\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle$ Vector-valued multivariable function "vector field"

Hessian is $\nabla^2 f = \begin{bmatrix} | & | & & | \\ \nabla \frac{\partial f}{\partial x_1} & \nabla \frac{\partial f}{\partial x_2} & \cdots & \nabla \frac{\partial f}{\partial x_n} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial f}{\partial x_2^2} & \\ \vdots & & \ddots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$

$(\nabla^2 f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

$n \times n$ matrix ↳ row i ↳ col j

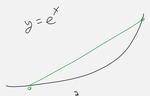
Matrix of functions ⇕ Function from $\mathbb{R}^n$ to Matrices. $\nabla^2 f : \mathbb{R}^n \to M_{n\times n}$

What can we say about symmetric matrices

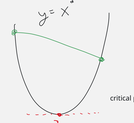
↳ (I) The function f is convex around a point a iff $\nabla^2 f$ is pos-semidefinite

Local min

Critical point min ⇔ "positive second derivative"



Convex, no minimum no critical points.



Convex one critical point is one minimum.

critical point of convex function is a minimum

Critical point

In one variable, $f: \mathbb{R} \to \mathbb{R}$ is C^2 and a is a critical point of f, Then $f''(a) > 0 \Rightarrow a$ is a local minimum.

$f''(a) > 0 \Rightarrow f$ is convex on a neighborhood of a

If f is convex around a and a is a critical point, then a is a local min

Pos 2nd derivative ⇒ Locally Convex

Convex + Critical Point ⇒ Min.

f is $C^2 \to \nabla f = \left\langle \frac{\partial f}{\partial x_i} \right\rangle_i$ $\quad \nabla^2 f = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j}$ Symmetric

Find Critical Points (at point p) $\nabla f = \vec{0}$ (i.e. all $\frac{\partial f}{\partial x_i} = 0$ at $\vec{p}$)

$\vec{u} = \nabla f(p)$ $S = \nabla^2 f(p)$ ↘ Spectral Theorem

New orthonormal basis $v_1, \dots, v_n$ where S is diagonal

Pos-Semidefinite:

S is pos-semidefinite $\iff x^T S x \ge 0 \ \forall x$

$\iff$ All eigenvals of S are $\lambda_1, \dots, \lambda_n$ $\quad \lambda_i \ge 0$

$[S]_B = Q^T S Q = D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ Signature $\to \begin{bmatrix} + & & \\ & + & \\ & & - \\ & & & 0 \end{bmatrix}$

$(S = QDQ^T)$

$Q^T S Q = Q^T Q D Q^T Q = D$

+ means $\lambda_i > 0$

− means $\lambda_i < 0$

0 means $\lambda_i = 0$

- positive semidefinite: + or 0 → f convex at $\vec{p}$
- positive definite: + only → f strictly convex at $\vec{p}$
- negative semidefinite: 0 or − → f concave (if convex) at $\vec{p}$
- negative definite: − only → f strictly concave (-f strictly convex) at $\vec{p}$

Convex + critical point is min

concave + critical point is max

Linear Algebra

Symmetric real matrix S $\quad$ (for any $\vec{p}$, $S = \nabla^2 f(\vec{p})$)

A real symmetric matrix

- Is its own transpose

Spectral Theorem:

- Always has n real eigenvalues (with multiplicity)
- Is orthogonally diagonalizable

↳ $S = QDQ^{-1}$ where Q is orthogonal matrix $(Q^{-1} = Q^T, \ Q^T Q = I$

$= QDQ^T$ columns of Q form an orthonormal basis)

Regular Diagonalizable: $S = PDP^{-1}$

where $D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ diagonal eigenvalues

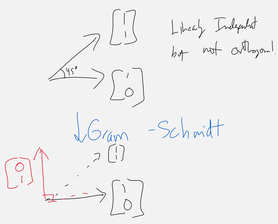
and $P = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{bmatrix}$ invertible eigenvector columns

where $Av_i = \lambda_i v_i$ and $v_1, \dots, v_n$ form basis.

If we can make an orthonormal eigenbasis, then Q ($= P$) is orthogonal

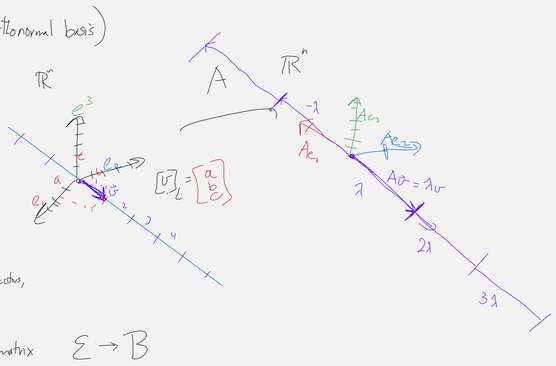
↳ $v_1, \dots, v_n$ are unit length and at right angles.

Change of basis $\mathcal{E} \to \mathcal{B}$ preserves length and angle.



Any matrix A has eigenvalue $\lambda \in \mathbb{R}$ if there is a nonzero vector $\vec{v} \in \mathbb{R}^n$ s.t. $A\vec{v} = \lambda \vec{v}$.

A fixes an axis (spanned by $\vec{v}$) and acts by scaling on that axis



If I can find n linearly independent eigenvectors $v_1, \dots, v_n$, then they form a basis of eigenvectors, called an eigenbasis $\mathcal{B} = (v_1, \dots, v_n)$

Take $P: \mathbb{R}^n \to \mathbb{R}^n$ as the change of basis matrix $\mathcal{E} \to \mathcal{B}$

$\mathcal{E} = (e_1, \dots, e_n)$ standard basis.

st. coord. $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ $\quad \vec{x} = ae_1 + be_2 + ce_3 = \alpha v_1 + \tilde{b} v_2 + \tilde{c} v_3$ $\quad \left[\vec{x}\right]_{\mathcal{B}} = \begin{bmatrix} \tilde{a} \\ \tilde{b} \\ \tilde{c} \end{bmatrix}$

We want $P \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \tilde{a} \\ \tilde{b} \\ \tilde{c} \end{bmatrix}$

$P\left([\vec{v}]_{\mathcal{E}} \right) = [\vec{v}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$v_1 = 1v_1 + 0v_2 + 0v_3$

The vectors are the same regardless of your basis

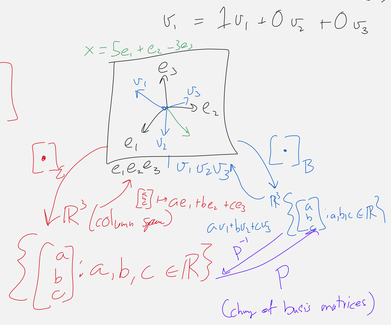
BUT

The columns are a "map" to a vector which are different in different bases.

Coordinate vector (map) in new basis (language)

$x = 5e_1 + e_2 - 3e_3$

$[x]_{\mathcal{E}} = \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix}$



$v_1 = ae_1 + be_2 + ce_3 = 1v_1 + 0v_2 + 0v_3$

$P \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

(coordinates of v in $\mathcal{E}$) (coordinates of v in $\mathcal{B}$) ⟷ "looks like standard basis"

$P^{-1} = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$